

FigureMy

```
+ Id: Guid
+ canva: Canvas
+ type: string
+ glob: Point
+ color: Brush
+ center_loc: Point
+ scale: double
+ angle: double
+ dop_angle: double
+ name: string
+ b_p1, b_p2: Point
+ border: Rectangle
+ CenterPoint: Ellipse
+ parent: FigureMy
+ children: ObservableCollection<FigureMy>
+ RW: RedWindow
~ dropping : bool
~ shapeStartPosition : Point
~ lastMousePosition : Point
```

```
Clone(part, parentCop): FigureMy
+ Insert(par): void
+ base_init(reinitial): void
+ Move(x, y): void
+ Move(): void
+ d_Move(dx, dy): void
+ d_Move_drag(dx, dy): void
+ setScale(new_scale): void
+ Draw(): void
+ Update_borders(): void
+ Edit(): void
+ Delete(): void
+ Deselect(): void
+ Select(): void
+ Rejection(): void
+ getGlobal(p): Point
+ getLocal(p): Point
```

SuperFigure : FigureMy

```
+ SuperFigure()
+ base_init(reinitial : bool) : void
+ Clone(part : FigureMy, parentCop : FigureMy) : FigureMy

+ setScale(new_scale : double) : void
+ AddFigure(figure : FigureMy) : void

+ Update_borders() : void

+ Rotate(newAngle : double, delta_dop_angle : double) : void
- RotatePoint(p : Point, center : Point, deltaAngle : double) : Point
```

AllFigures : SuperFigure

```
+ AllFigures()

+ Delete() : void
+ Select() : void
+ Deselect() : void
```

HandlePolygon : FigureMy

```
- points : List<Point>
- currentLine : Line
- lines : List<Line>

- canvas : Canvas
- isDrawing : bool

- lineEditWindow : HandleLineEditWindow
```

```
+ HandlePolygon()

+ Start() : void
- CreateLine(p1 : Point, p2 : Point) : Line

// Пересечения
- Cross(a : Point, b : Point, c : Point) : double
- OnSegment(a : Point, b : Point, p : Point) : bool
- LinesIntersect(a1 : Point, a2 : Point, b1 : Point, b2 : Point) : bool
- HasIntersection(last : Point, newPoint : Point) : bool

// События
- Canvas_MouseMove(sender : object, e : MouseEventArgs) : void
- Canvas_MouseLeftButtonDown(sender : object, e : MouseButtonEventArgs) : void
- Canvas_MouseRightButtonDown(sender : object, e : MouseButtonEventArgs) : void
- Canvas_KeyDown(sender : object, e : KeyEventArgs) : void

// Логика линии
- UpdateLineFromWindow(length : double, angle : double, x2 : double, y2 : double, fromWindow : bool) : void
- ApplyLineFromWindow(length : double, angle : double, x2 : double, y2 : double) : void

- DetachEvents() : void
```

Circle : FigureMy

PolygonMy : FigureMy

```
+ sides : List<Side>
```

